SOFTWARE ENGINEERING

Lab 3: Architectural Pattern Justification

Problem Statement #31 — Multi-Vendor Artisan E-Commerce Marketplace

Architecture Selection: We chose Microservices Architecture for the Multi-Vendor Artisan E-Commerce Marketplace System.

Reason 1 — Independent Scaling

NFR-001 requires the product catalog to render high-resolution vendor images with CDN caching at under 500 ms, which means the Catalog Service sees very high, bursty read traffic from browsing shoppers. The Payment & Payout Service, in contrast, handles lower-volume but strongly consistent transactions (FR-001's split-payout logic). Microservices let each service scale on its own hardware/replica count and its own schedule, instead of a monolith forcing the whole system to scale together whenever catalog traffic spikes.

Reason 2 — Fault Isolation Across Independent Vendors

The platform onboards many independent artisan vendors, each managing their own storefront and catalog. If the Vendor Account Service or a payout job slows down or fails for one vendor, a microservices split ensures the Catalog and Shopping Cart services keep functioning normally for shoppers, rather than an issue in one vendor's data bringing down browsing and checkout platform-wide, as could happen in a tightly coupled monolith or layered design.

Security Advantage

Splitting the Payment & Payout Service into its own microservice allows it to be deployed in a restricted, separately-secured network segment with a minimal, tightly controlled set of interfaces (handling commission deductions and vendor payouts). This limits the blast radius: a vulnerability in the public-facing, image-heavy Catalog Service cannot directly reach payment credentials, vendor bank details, or payout logic, since that sensitive data never leaves the isolated payment service boundary.

Performance Benefit

Because the Catalog Service is deployed independently, it can apply its own CDN edge caching, image-optimisation pipeline, and horizontally-scaled read replicas purely for browsing traffic, meeting the sub-500 ms NFR-001 target without being slowed down by the heavier, synchronous work of order creation and payment splitting that happens in the Order Management and Payment & Payout services.